

GIRRAWEE HIGH SCHOOL
YEAR 11 YEARLY EXAMINATION

2005

MATHEMATICS

2 UNIT

*Time allowed – Two Hours
(Plus 5 minutes reading time)*

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question.
- Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Start each question on a new page.

Question 1. (21 marks)**Marks**

(a) Find the exact value of:

(i) $\frac{1}{4}(x^3 - x^2 + 4)$ when $x = -2$

1

(ii) $49^{-\frac{1}{2}} \times 8^{\frac{1}{3}}$

2

(b) Rationalise the denominator and express in simplest surd form.

3

$$\frac{2\sqrt{3} + \sqrt{2}}{\sqrt{3} + \sqrt{2}}$$

(c) Solve the equation $\frac{5x+1}{3x+1} = \frac{5x+2}{3x-2}$

3

(d) Simplify $\frac{x+1}{x^2-x} - \frac{x-1}{x^2+x}$

3

(e) Factorise fully $4x^3 - 32$

2

(f) Express $0.\dot{2}\dot{3}$ as a fraction in simplest form.

2

(g) Find all the solutions for:

(i) $|2x - 1| = 3x + 5$

3

(ii) $|x - 3| < 2$

2

Question 2. (26 marks) (Start on a new page)

(a) Find the derivative of:

(i) $x^5 + 4x^2 - 7$ 2

(ii) $\frac{2x+1}{3x-2}$ 3

(iii) $(2x^2 - 3)^5$ 2

(iv) $\sqrt{x^3}$ 2

(b) Find, from first principles, the slope of the tangent to the curve $y = x^2 + 2$ at the point $(-1, 3)$. 3

(c) For the curve $y = ax^2 + bx + c$ where a, b, c are constants, 3
it is given that when $x = 1, y = 1$ and $\frac{dy}{dx} = 1$. Show that $a = c$.

(d) Evaluate:

(i) $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$ 2

(ii) $\lim_{x \rightarrow \infty} \frac{2x^2 + 5}{x^2 + 3x}$ 2

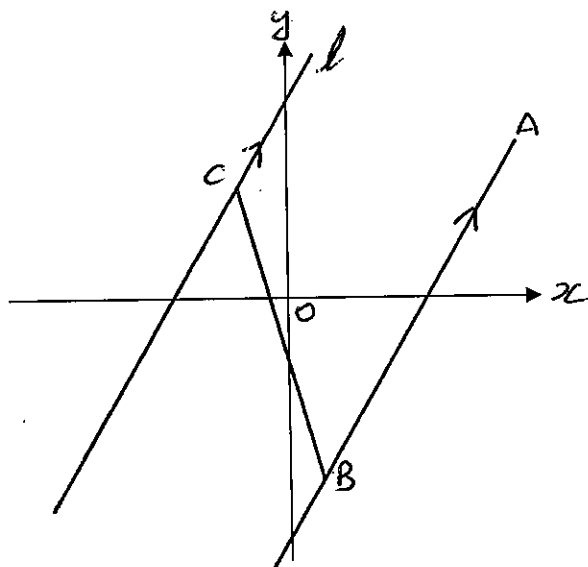
(e) Find the equation of the normal to the curve $y = x^3 - 3x^2 - 9x + 30$ at the point $(2, 8)$. 4

(f) Find the coordinates of the points where the line $y = 30 - 9x$ intersects the curve $y = x^3 - 3x^2 - 9x + 30$ 3

Question 3. (18 marks)

(Start on a new page)

(a)



The line l passes through $C(-1, 2)$ and has equation $y = 2x + 4$.

The point B has coordinates $(1, -6)$ and the line AB is parallel to l .

- | | | |
|-------|--|---|
| (i) | Find the length of the interval BC . | 2 |
| (ii) | Find the mid-point of the interval BC . | 2 |
| (iii) | Write down the slope of the line l and use your calculator to find the acute angle l makes with the x -axis. Give your answer to the nearest degree. | 3 |
| (iv) | Show that AB has equation $y = 2x - 8$. | 2 |
| (v) | If P is a point which lies on AB , and on the line $y = 2$, find the coordinates of P . | 1 |
| (vi) | Find the length of PC . | 1 |
| (vii) | Find the area of triangle PBC . | 2 |
| (b) | Two perpendicular lines $3x + 2y = 5$ and $4x + ay = b$ intersect in the point $(1, 1)$. Find the values of a and b . | 3 |
| (c) | Find the perpendicular distance from the point $A(5, 7)$ to the line $3x + 4y - 18 = 0$. | 2 |

Question 4. (21 marks) (Start on a new page)

- (a) Find the exact value of the following without using a calculator:

(i) $\tan 150^\circ$. 1

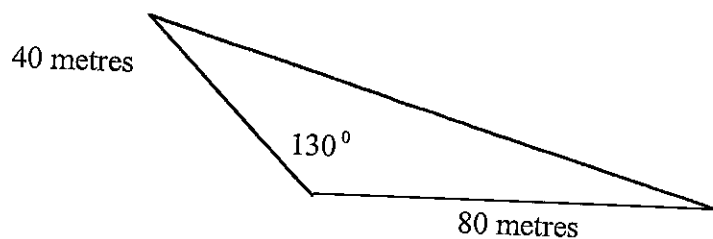
(ii) $\sin^2 30^\circ + \cos^2 30^\circ$. 2

(b) If $\sin \theta = -\frac{\sqrt{3}}{2}$, find all the values of θ for $0^\circ \leq \theta \leq 360^\circ$. 2

(c) Given θ is in the second quadrant and $\cos \theta = -\frac{1}{2}$,
find the exact value for $\sin \theta + \tan \theta$ 3

(d) Simplify fully $\frac{1}{1 - \cos \theta} + \frac{1}{1 + \cos \theta}$ 3

(e)

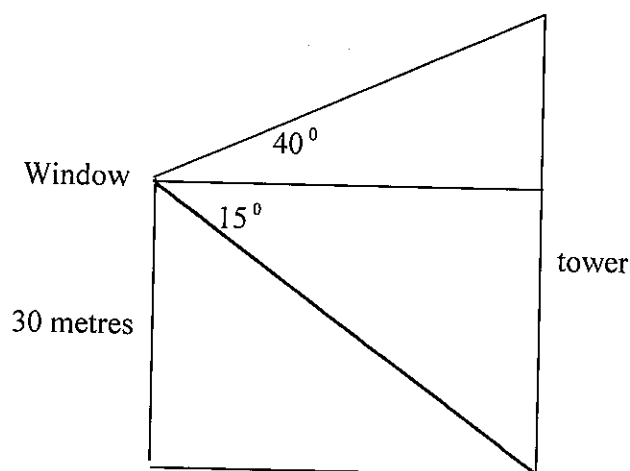


A field is triangular, with two sides of 80 metres and 40 metres enclosing an angle of 130° .

- (i) Calculate the area of the field. 2

- (ii) Use the cosine rule to calculate the length of the third side to the nearest metre. 3

(f)



From a window 30m above the ground, the angle of elevation of the top of a church tower is found to be 40° , and the angle of depression of the bottom of the tower to be 15° .

- (i) Find the horizontal distance from the observer to the tower. 2
- (ii) Find the height of the tower. 3

Question 5. (17 marks) (Start on a new page)

- (a) If α and β are the roots of the equation $2x^2 - 5x - 1 = 0$, find an expression for:
 - (i) $\alpha + \beta$. 1
 - (ii) $\alpha\beta$. 1
 - (iii) $\alpha^2 + \beta^2$. 2
- (b) For what values of k does the expression $x^2 + (k-3)x + k = 0$ have real roots? 3
- (c) For what values of k is the expression $x^2 + (k-3)x + k$ positive for all values of x ? 3
- (d) Solve $4^x - 3(2^x) + 2 = 0$. 4
- (e) Express $4x^2 + x - 1$ in the form $A + B(x+1) + Cx(x+1)$. 3

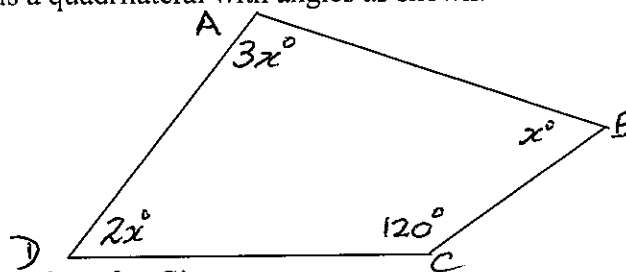
Question 6. (18 marks) (Start on a new page)

- (a) A function is defined as: $f(x) = \begin{cases} 2x-1 & \text{for } 0 \leq x \leq 3 \\ \frac{1}{3}x+4 & \text{for } 3 < x \leq 5 \end{cases}$
- (i) Find the domain of the function. 1
- (ii) Find the range of the function. 1
- (iii) Find the value of $f(4) - f(2)$. 2
- (b) For the function $y = \sqrt{9 - x^2}$,
- (i) Find the domain. 2
- (ii) Find the range. 1
- (c) On the same set of axes, sketch the graph of:
- (i) $y = 2 - x^2$. 2
- (ii) $y = x$. 1
- (iii) Shade the region where $y \leq 2 - x^2$ and $y \geq x$. 2
- (d) Draw a sketch graph for each of the following functions:
- (i) $xy = -9$. 2
- (ii) $y = 2^x$. 2
- (iii) $x^2 + y^2 = 4$ 2

Question 7. (18 marks)

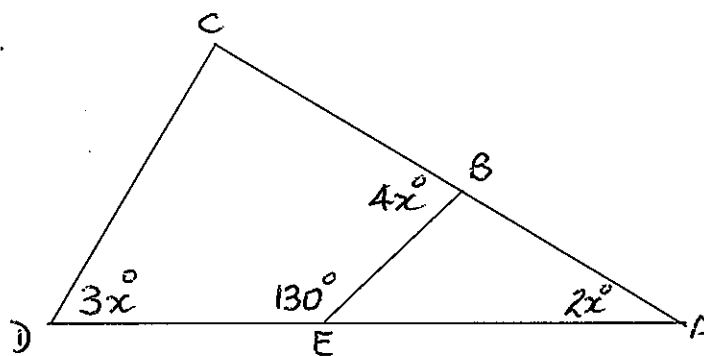
(Start on a new page)

- (a) ABCD is a quadrilateral with angles as shown. 3



Find the value of x . Give reasons.

(b)



In $\triangle ACD$, the points B and E lie on the sides AC and AD respectively.

(i) Find the value of x.

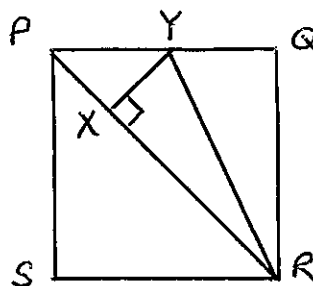
3

(ii) Hence find $\angle ACD$ in degrees.

2

Give reasons for your answers.

(c)



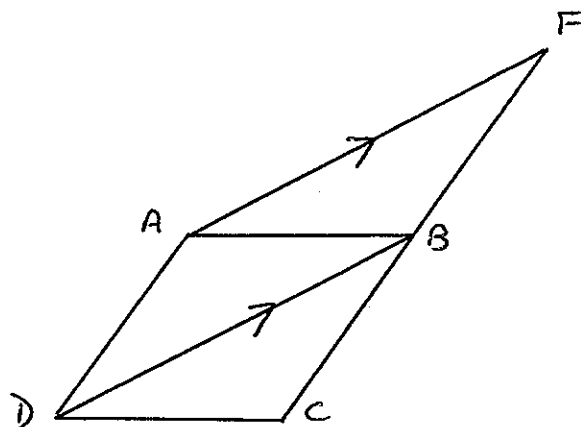
PQRS is a square. YR bisects $\angle PRQ$.

X lies on PR such that $XY \perp PR$.

Prove that $\triangle YQR$ and $\triangle YXR$ are congruent.

4

(d)



ABCD is a rhombus in which $AD = 6\text{cm}$ and $BD = 9\text{cm}$.

$FA \parallel BD$ and the points C, B, F are collinear.

Find, giving reasons:

(i) the area of rhombus ABCD, correct to 2 decimal places.

3

(ii) the area of the figure AFCD, correct to 2 decimal places.

3

2005 Year 11 2 Unit Solutions.

Question 1 (21 marks)

$$\begin{aligned} (a) (i) & \frac{1}{4}((-2)^3 - (-2)^2 + 4) \\ &= \frac{1}{4}(-8 - 4 + 4) \\ &= \frac{1}{4} \times -8 \\ &= -2 \end{aligned} \quad (1)$$

$$\begin{aligned} (ii) & 49^{-\frac{1}{2}} \times 8^{\frac{1}{3}} \\ &= \frac{1}{\sqrt{49}} \times \sqrt[3]{8} \\ &= \frac{2}{7} \end{aligned} \quad (2)$$

$$\begin{aligned} (b) & \frac{2\sqrt{3} + \sqrt{2}}{\sqrt{3} + \sqrt{2}} \times \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} - \sqrt{2}} \\ &= \frac{6 - 2\sqrt{6} + \sqrt{6} - 2}{3 - 2} \\ &= 4 - \sqrt{6} \end{aligned} \quad (3)$$

$$\begin{aligned} (c) & \frac{5x+1}{3x+1} = \frac{5x+2}{3x-2} \\ (5x+1)(3x-2) &= (5x+2)(3x+1) \\ 15x^2 + 3x - 10x - 2 &= 15x^2 + 6x + 5x + 2 \\ -7x - 2 &= 11x + 2 \\ -4 &= 18x \\ x &= -\frac{2}{9} \end{aligned} \quad (3)$$

$$\begin{aligned} (d) & \frac{x+1}{x^2-x} - \frac{x-1}{x^2+x} \\ &= \frac{(x+1)^2 - (x-1)^2}{x(x-1)(x+1)} \\ &= \frac{(x^2+2x+1) - (x^2-2x+1)}{x(x-1)(x+1)} \\ &= \frac{4x}{x(x-1)(x+1)} \\ &= \frac{4}{(x-1)(x+1)} \end{aligned} \quad (2)$$

$$\begin{aligned} (e) & 4x^3 - 32 \\ &= 4(x^3 - 8) \\ &= 4(x-2)(x^2+2x+4) \end{aligned} \quad (2)$$

$$\begin{aligned} (f) & \text{Let } x = 0.2323 \dots \\ 100x &= 23.2323 \\ 99x &= 23 \\ x &= \frac{23}{99} \end{aligned} \quad (2)$$

$$\begin{aligned} (g) (i) & |2x-1| = 3x+5 \\ 2x-1 &= 3x+5 \quad \text{or} \quad -(2x-1) = 3x+5 \\ -6 &= x & -2x+1 &= 3x+5 \quad (3) \\ |13| &\neq -13 & -4 &= 5x \\ x &= -6 \text{ is NOT a solution} & -4 &= x \\ & & 5 & \end{aligned}$$

$$\begin{aligned} (ii) & |x-3| < 2 \\ -2 & < x-3 < 2 \\ 1 & < x < 5 \end{aligned} \quad (2)$$

Question 2 (26 marks)

(a) (i) $\frac{d}{dx}(x^5 + 4x^2 - 7)$

$$= 5x^4 + 8x \quad (2)$$

(ii) $\frac{d}{dx}\left(\frac{2x+1}{3x-2}\right)$

$$= \frac{(3x-2) \times 2 - 3(2x+1)}{(3x-2)^2}$$

$$= \frac{6x-4-6x-3}{(3x-2)^2} \quad (3)$$

$$= -\frac{7}{(3x-2)^2}$$

(iii) $\frac{d}{dx}[(2x^2-3)^5]$

$$= 5(2x^2-3)^4 \times 4x$$

$$= 20x(2x^2-3)^4 \quad (2)$$

(iv) $\frac{d}{dx}(\sqrt{x^3})$

$$= \frac{d}{dx}(x^{3/2})$$

$$= \frac{3}{2}x^{1/2} \quad (2)$$

$$= \frac{3}{2}\sqrt{x}$$

(d) (i) $\lim_{x \rightarrow 5} \frac{x^2-25}{x-5}$

$$= \lim_{x \rightarrow 5} \frac{x+5}{1}$$

$$= 10 \quad (2)$$

(ii) $\lim_{x \rightarrow \infty} \frac{2x^2+5}{x^2+3x}$

$$= \lim_{x \rightarrow \infty} \frac{\frac{2x^2}{x^2} + \frac{5}{x^2}}{\frac{x^2}{x^2} + \frac{3x}{x^2}}$$

$$= \lim_{x \rightarrow \infty} \frac{2 + \frac{5}{x^2}}{1 + \frac{3}{x}}$$

$$= 2 \quad (2)$$

(b) $\frac{[(x+h)^2-12] - [x^2-12]}{h}$

$$= \frac{x^2 + 2xh + h^2 - 12 - x^2 + 12}{h}$$

$$= \frac{2xh + h^2}{h}$$

$$= 2x + h$$

$$\lim_{h \rightarrow 0} (2x + h)$$

$$= 2x$$

At $x = -1$, $\frac{dy}{dx} = -2$.

Slope of tangent $= -2 \quad (3)$

(c) $y = ax^2 + bx + c \quad \text{--- (1)}$

$$\frac{dy}{dx} = 2ax + b \quad \text{--- (2)}$$

Sub $x=1, y=1$ in (1)

$$1 = a + b + c$$

Sub $x=1, \frac{dy}{dx} = 1$ in (2)

$$1 = 2a + b$$

By subtraction (3)

$$a - c = 0$$

$$a = c$$

(e) $y = x^3 - 3x^2 - 9x + 30$

$$\frac{dy}{dx} = 3x^2 - 6x - 9$$

At $x=2$, $\frac{dy}{dx} = 3 \times 4 - 6 \times 2 - 9 = -9$

Gradient of normal $= \frac{1}{9}$

Equation of normal:

$$(y-8) = \frac{1}{9}(x-2)$$

$$9y - 72 = x - 2$$

$$x - 9y + 70 = 0. \quad (4)$$

(f) $x^3 - 3x^2 - 9x + 30 = 30 - 9x$

$$x^3 - 3x^2 = 0$$

$$x^2(x-3) = 0$$

$$x = 0 \text{ or } x = 3$$

$$y = 30 \text{ or } y = 3$$

Question 3 (18 marks)

(a) (i) $BC = \sqrt{(1+1)^2 + (-6-2)^2}$
 $= \sqrt{2^2 + (-8)^2}$
 $= \sqrt{4+64}$
 $= \sqrt{68} \text{ units} \quad (2)$

(ii) Midpoint of $BC = \left(\frac{-1+1}{2}, \frac{2-6}{2} \right)$
 $= \left(\frac{0}{2}, \frac{-4}{2} \right)$
 $= (0, -2) \quad (2)$

(iii) Slope = 2
 $\tan \theta = 2$
 $\theta = 63^\circ \quad (3)$

(iv) $AB \parallel l$
 $\therefore \text{Gradient } AB = 2$
 $B(1, -6) \text{ lies on } AB$

Equation AB
 $(y+6) = 2(x-1)$
 $y = 2x - 2 - 6$
 $y = 2x - 8 \quad (2)$

(v) If $y=2$, $2 = 2x - 8$
 $10 = 2x$
 $5 = x$

(vi) Length $PC = \sqrt{(5+1)^2 + (2-2)^2}$
 $= 6 \text{ units} \quad (1)$

P is the point $(5, 2) \quad (1)$

(b) $3x + 2y = 5$

$4x + ay = b$

(c) Perpendicular distance

$2y = -3x + 5$

$ay = -4x + b$

$y = -\frac{3}{2}x + \frac{5}{2}$

$y = -\frac{4}{a}x + \frac{b}{a}$

$= \frac{|3 \times 5 + 4 \times 7 - 18|}{\sqrt{3^2 + 4^2}}$

For perpendicular lines: $-\frac{3}{2}x - \frac{4}{a} = -1$ $= \frac{|15 + 28 - 18|}{\sqrt{25}}$

$\frac{6}{a} = -1$

$a = -6$

$= \frac{|25|}{5}$

Sub $x=1, y=1, a=-6$ in

$4x + ay = b$

$4 - 6 = b$

$-2 = b \quad (3)$

$= 5 \text{ units} \quad (2)$

Question 4 (21 marks)

(a) (i) $\tan 150^\circ = -\frac{1}{\sqrt{3}}$ (1)

(b) $\sin \theta = -\frac{\sqrt{3}}{2}$
 $\theta = (180-60)^\circ \text{ or } (360-60)^\circ$
 $= 240^\circ \text{ or } 300^\circ$ (2)

(ii) $\sin^2 30^\circ + \cos^2 30^\circ$
 $= \left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2$ (2)
 $= \frac{1}{4} + \frac{3}{4} = 1$

(c) $\cos \theta = -\frac{1}{2}$
 $\theta = (180-60)^\circ = 120^\circ$

(d) $\frac{1}{1-\cos \theta} + \frac{1}{1+\cos \theta}$
 $= \frac{1+\cos \theta + 1-\cos \theta}{1-\cos^2 \theta}$
 $= \frac{2}{\sin^2 \theta}$ (3)
 $= 2 \operatorname{cosec}^2 \theta$

$\sin 120^\circ + \tan 120^\circ$
 $= \frac{\sqrt{3}}{2} + \left(-\frac{\sqrt{3}}{1}\right)$
 $= \frac{-\sqrt{3}}{2}$ (3)

(e) (i) Area = $\frac{1}{2} \times 40 \times 80 \times \sin 130^\circ$
 $= 1226 \text{ m}^2 \text{ (to nearest m}^2\text{)} (2)$

(f) (i) $\tan 15^\circ = \frac{\text{distance}}{30}$
 $\text{distance} = \frac{30}{\tan 15^\circ}$ (2)

(ii) $d^2 = 40^2 + 80^2 - 2 \times 40 \times 80 \times \cos 130^\circ$
 $= 1600 + 6400 - 413.84$
 $= 1213.84$
 $d = \sqrt{1213.84}$ (3)
 $= 110 \text{ m (to nearest metre)}$

(ii) $\tan 40^\circ = \frac{h}{112}$
 $h = 112 \tan 40^\circ$ (3)
 $= 94 \text{ m to nearest m.}$

Height of tower = $94 + 30$
 $= 124 \text{ m.}$

Question 5 (17 marks)

(a) (i) $\alpha + \beta = -\frac{-5}{2} = \frac{5}{2}$ (1)

(ii) $\alpha\beta = -\frac{1}{2}$ (1)

(iii) $\alpha^2 + \beta^2$
 $= (\alpha + \beta)^2 - 2\alpha\beta$
 $= \left(\frac{5}{2}\right)^2 - 2\left(-\frac{1}{2}\right)$
 $= \frac{25}{4} + 1$
 $= \frac{29}{4}$

$= 7\frac{1}{4}$

(2)

(b) $\Delta \geq 0$

$(k-3)^2 - 4 \times 1 \times k \geq 0$

$k^2 - 6k + 9 - 4k \geq 0$

$k^2 - 10k + 9 \geq 0$

$(k-9)(k-1) \geq 0$

$k \leq 1, k \leq 9$ OR $k \geq 1, k \geq 9$

$k \leq 1$ $k \geq 9$ (3)

(c) $\Delta < 0$

$k^2 - 10k + 9 < 0$

$(k-9)(k-1) < 0$

(3)

$k-9 > 0$ and $k-1 < 0$ OR $k-9 < 0$ and $k-1 > 0$

$k > 9$ and $k < 1$ $k < 9$ and $k > 1$

No solutions $\therefore 1 < k < 9$

(d) $4^x - 3(2^x) + 2 = 0$

Let $u = 2^x$

$u^2 - 3u + 2 = 0$

$(u-2)(u-1) = 0$

$\therefore u = 2$ OR $u = 1$

$2^x = 2$ $2^x = 1$ (4)

$x = 1$ $x = 0$

(e) $4x^2 + x - 1 \equiv A + B(x+1) + Cx(x+1)$

$\equiv A + Bx + B + Cx^2 + Cx$

$\equiv (A+B) + (B+C)x + Cx^2$

$\therefore C = 4$

$B + C = 1$

$\therefore B = -3$

$A + B = -1$

$\therefore A = 2$ (3)

$4x^2 + x - 1 \equiv 2 - 3(x+1) + 4x(x+1)$

Question 6 (18 marks)

(a) (i) Domain $0 \leq x \leq 5$ (1)

(ii) Range When $x=0$, $y=-1$

When $x=5$, $y=\frac{5}{3}+4=5\frac{2}{3}$

Range $-1 \leq y \leq 5\frac{2}{3}$ (1)

(iii) $f(4) - f(2)$

$= (\frac{1}{3} \times 4 + 4) - (2 \times 2 - 1)$

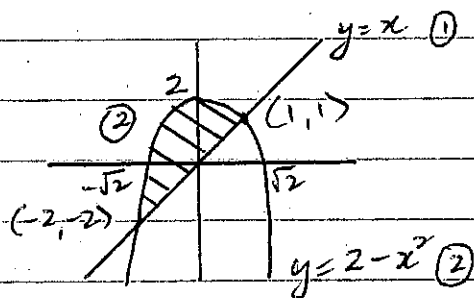
$= 5\frac{1}{3} - 3$

$= 2\frac{1}{3}$ (2)

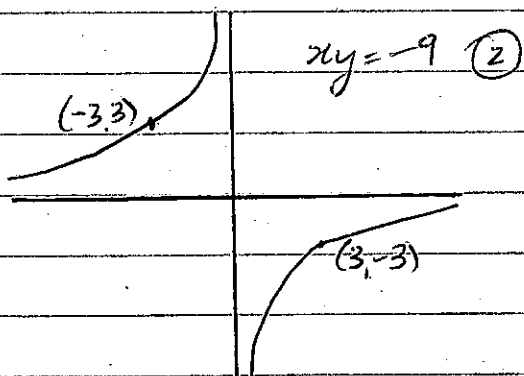
(b) (i) Domain $-3 \leq x \leq 3$ (2)

(ii) Range $0 \leq y \leq 3$ (1)

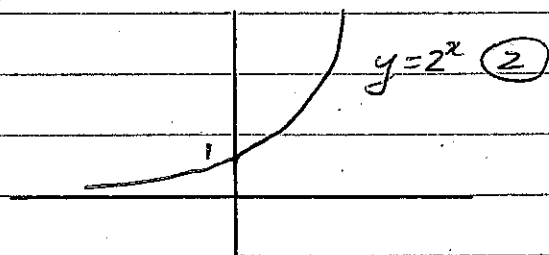
(c)



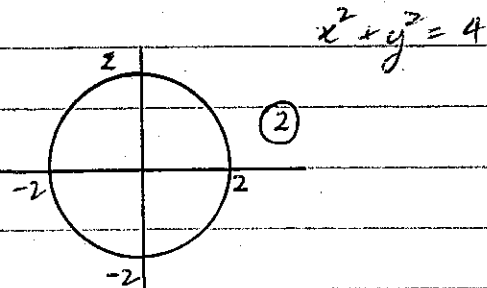
(d) (i)



(ii)



(iii)



Question 7 (18 marks)

(a) $2x + 3x + x + 120 = 360$ (angle sum of a quadrilateral)

$$6x + 120 = 360$$

$$6x = 240$$

$$x = 40 \quad (3)$$

(b) (i) $\angle EBA = 180 - 4x$ (supplementary angles)

$\angle DEB = \angle EBA + \angle BAE$ (exterior angle of triangle is the sum of the two interior opposite angles)

$$130 = 180 - 4x + 2x$$

$$130 = 180 - 2x$$

$$2x = 50$$

$$x = 25 \quad (3)$$

(ii) $2x + 3x + \angle ACD = 180$ (angle sum of a triangle)

$$5x + \angle ACD = 180$$

$$125 + \angle ACD = 180$$

$$\angle ACD = 55^\circ \quad (2)$$

(c) In $\triangle YQR$ and $\triangle YXR$

$$\angle YQR = \angle YXR \text{ (angles of a square, data)}$$

$$\angle YRQ = \angle YRX \text{ (YR bisects } \angle PRQ)$$

YR is common

$$\therefore \triangle YQR \equiv \triangle YXR \text{ (AAS)} \quad (4)$$

(d) (i) Let AC meet BD in E.

$$DE^2 + EA^2 = AD^2 \text{ (diagonals of a rhombus bisect each other at right angles)}$$

$$4.5^2 + EA^2 = 6^2$$

$$EA^2 = 6^2 - 4.5^2$$

$$= 15.75$$

$$EA = \sqrt{15.75}$$

$$\text{Area of rhombus} = \frac{1}{2} \times 9 \times 2 \times \sqrt{15.75}$$

$$= 35.71 \text{ cm}^2 \quad (3)$$

(ii) $AD \parallel CB$ (opposite sides of rhombus)

$AF \parallel BD$ (data)

$\therefore AFBD$ is a parallelogram.

AB divides the parallelogram into

two congruent triangles. (3)

$$\therefore \text{Area } AFC = \frac{3}{2} \times 9 \times \sqrt{15.75}$$

$$= 53.58 \text{ cm}^2$$